1 Case Study 2: Institutional Benchmarking

1.1 Objective

Compare citation summaries, fractional credits and open-access metadata in capped publication samples from five universities. These descriptive samples do not measure total productivity or field-normalised impact.

1.2 Setup

```
library(tidyverse)
library(openalexR)
library(glue)
library(gt)

set.seed(20260509)

source(here::here("R", "api_helpers.R"))
source(here::here("R", "utils.R"))
source(here::here("R", "sci_palette.R"))
source(here::here("R", "institutional_samples.R"))
```

1.3 Data acquisition

Institution IDs were checked against OpenAlex on 30 October 2026, including the University of São Paulo record. Each request retrieves at most 300 articles; these are samples, not output totals. A failed or empty request stops the comparison instead of silently dropping an institution.

```
institutions <- tribble(
  ~short_name,    ~oa_id,
  "MIT",          "I63966007",
  "Oxford",       "I40120149",
  "ETH Zurich",   "I35440088",
  "U São Paulo",  "I17974374",
  "U Cape Town",  "I157614274"
)
```

```
fetch_works <- function(inst_id) {
  res <- fetch_openalex(
    entity = "works",
    authorships.institutions.id = inst_id,
    from_publication_date = "2020-01-01",
    to_publication_date = "2022-12-31",
    type = "article",
    options = list(sample = 300, seed = 42)
  )
  if (!is.data.frame(res) || !nrow(res)) {
    stop(glue("No usable works returned for {inst_id}; check the request."))
  }
  res
}
```

```
inst_data <- institutions |>
  mutate(works = map(oa_id, fetch_works))
```

1.4 Publication volume

```
inst_data |>
  mutate(n = map_int(works, nrow)) |>
  ggplot(aes(x = n, y = reorder(short_name, n))) +
  geom_col(fill = palette_sci(1)) +
  labs(x = "Sampled articles", y = NULL) +
  theme_sci()
```

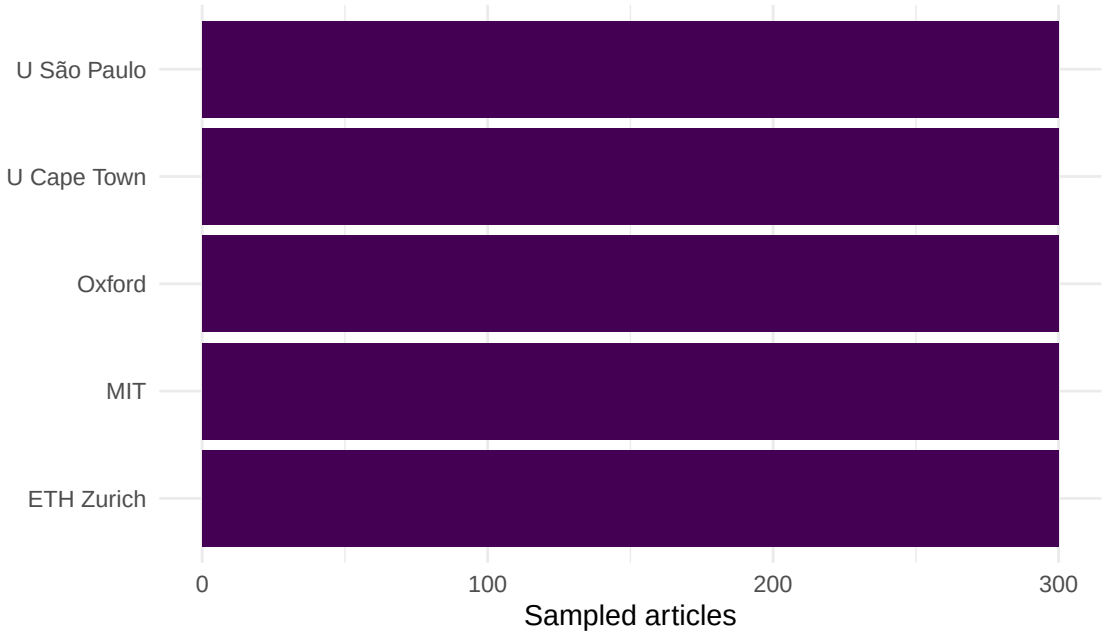


Figure 1: Sampled publication counts by institution.

1.5 Citation summaries

As in Chapter 13, the citation ratio uses a common year reference from the union of the five samples, de-duplicated by work ID. It adjusts for year only, not field mix or a common citation window. Missing citation counts and undefined ratios are excluded with their usable denominators shown.

```
citation_reference <- sci_pooled_year_reference(inst_data$works)
impact <- inst_data |>
  mutate(summary = map(works, sci_sample_citation_summary,
                       reference = citation_reference)) |>
  unnest(summary) |>
  select(short_name, n_sampled, n_citation_observed, mean_cites,
         median_cites, n_ratio_observed, pooled_year_ratio) |>
```

short_name	n_sampled	n_citation_observed	mean_cites	median_cites	n_ratio_observe
ETH Zurich	300	300	45.32	25.00	30
MIT	300	300	47.54	24.00	30
Oxford	300	300	46.42	19.00	30
U Cape Town	300	300	29.21	12.00	30
U São Paulo	300	300	13.51	6.00	30

short_name	full_count	affiliation_covered	frac_count	ratio
MIT	300	256	NA	NA
Oxford	300	263	NA	NA
ETH Zurich	300	276	NA	NA
U São Paulo	300	254	NA	NA
U Cape Town	300	249	NA	NA

```
  arrange(short_name)

impact |>
  gt() |>
  fmt_number(columns = c(mean_cites, median_cites, pooled_year_ratio), decimals = 2)
```

1.6 Full vs. fractional counting

Count each distinct affiliation once per work and divide its credit equally. Incomplete affiliations make the fractional total unavailable; the coverage column shows how many sampled works have usable data. This is bibliometric credit, not an estimate of each institution's contribution.

```
counting <- inst_data |>
  mutate(counts = map2(works, oa_id, sci_institution_sample_counts)) |>
  unnest(counts) |>
  select(short_name, full_count, affiliation_covered, frac_count) |>
  mutate(ratio = round(frac_count / full_count, 2))

counting |> gt()
```

```
counting |>
  pivot_longer(c(full_count, frac_count), names_to = "method", values_to = "count") |>
  mutate(method = recode(method, full_count = "Full", frac_count = "Fractional")) |>
  ggplot(aes(x = count, y = reorder(short_name, count), fill = method)) +
  geom_col(position = "dodge", width = 0.7) +
  scale_fill_manual(values = palette_sci(2)) +
  labs(x = "Count", y = NULL, fill = "Method") +
  theme_sci()
```

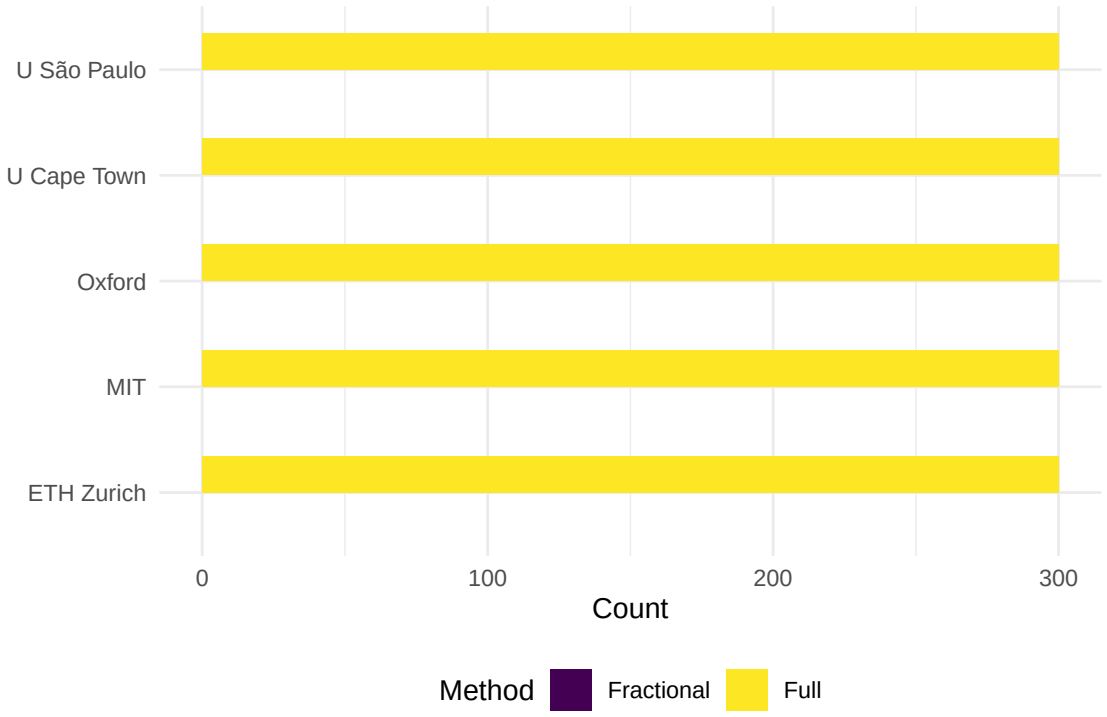


Figure 2: Full and fractional credits within the retrieved samples.

1.7 OA rates

```
inst_data |>
  mutate(oa_rate = map_dbl(works, \(w) mean(w$oa_status != "closed", na.rm = TRUE))) |>
  ggplot(aes(x = oa_rate, y = reorder(short_name, oa_rate))) +
  geom_col(fill = palette_sci(1)) +
  scale_x_continuous(labels = scales::percent) +
  labs(x = "OA rate", y = NULL) +
  theme_sci()
```

1.8 Reading the results

1. **Sample size**: Counts are capped by the request and cannot rank institutional productivity.
2. **Citation summaries**: Means and pooled sample/year ratios describe the retrieved records. Neither is field-normalised; an h-index would also depend on sample size and is omitted here.
3. **Fractional credit**: A lower fractional/full ratio means less credit is retained under this equal-institution counting rule, subject to complete affiliation coverage. It does not by itself rank the average affiliation count or collaboration quality.
4. **OA metadata**: Differences in observed OA shares do not establish the effects of national mandates or institutional policies.
5. **No single ranking**: Report coverage, sample design and indicator definitions alongside any comparison [@hicks2015].

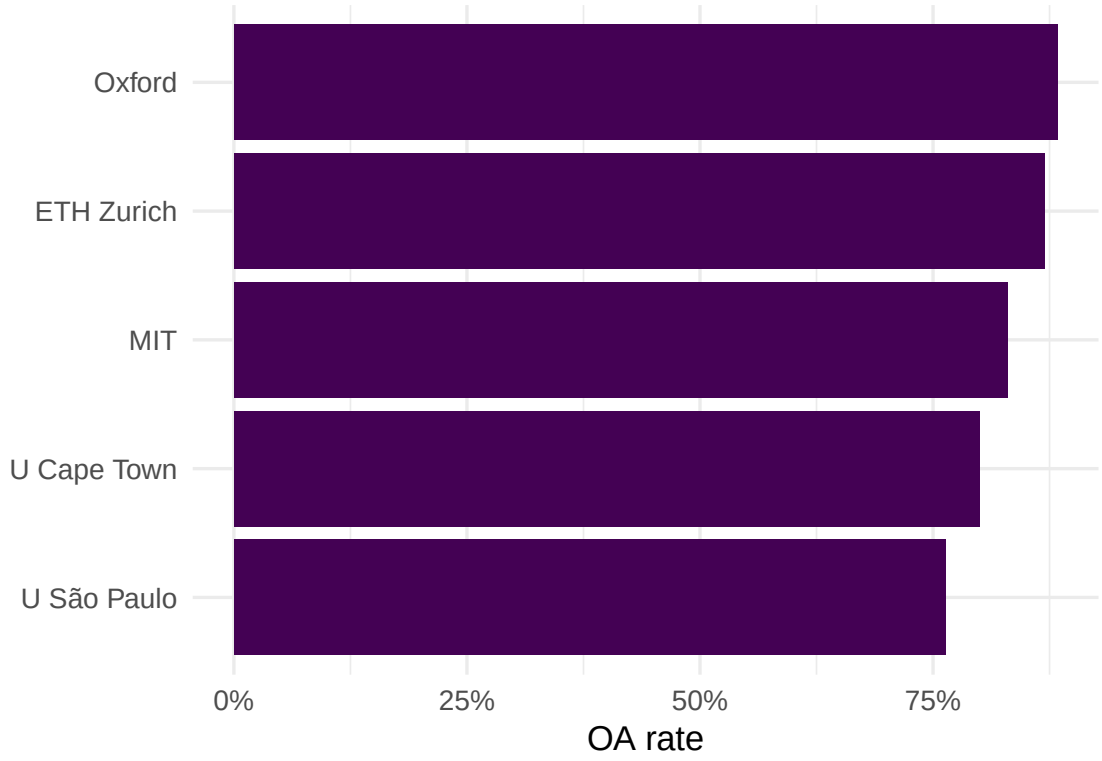


Figure 3: Open-access share among sampled articles with observed OA status.

1.9 Lessons learned

- Treat these outputs as illustrative sampled comparisons, not population rankings.
- Field mix and differing citation windows confound comparisons.
- Responsible benchmarking requires a justified reference set, transparent counting rules, missing-data coverage and uncertainty analysis.